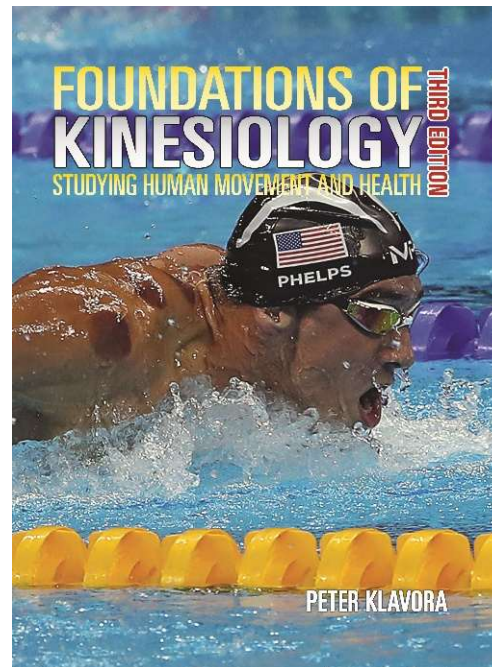


FOUNDATIONS OF KINESIOLOGY:

STUDYING HUMAN MOVEMENT & HEALTH

3RD EDITION



Kinesiology Books Publisher



CHAPTER 3

What Is Anatomy?

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The Language of Anatomy



- Anatomical Position
- Directional Terms
- Body Planes
- Movements

Musculoskeletal System



- Human Skeleton
- Types of Muscle
- Joints



3.1 THE LANGUAGE OF ANATOMY

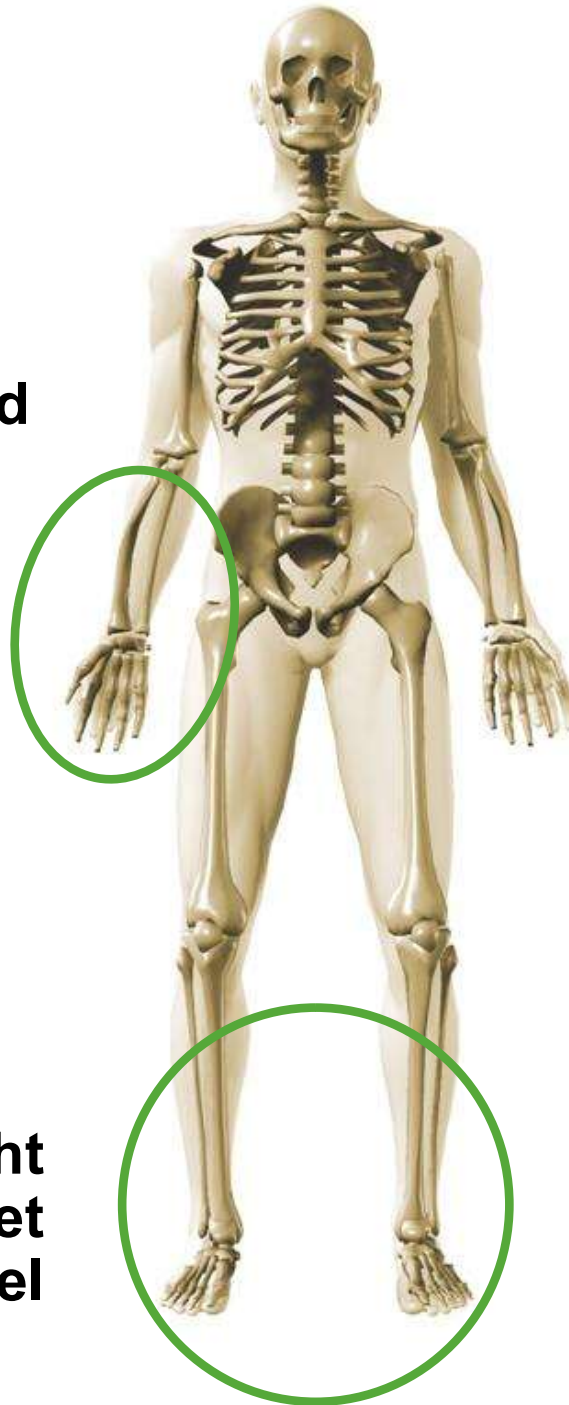
- **Anatomical Position**
- **Directional Terms**
- **Body Planes**
- **Movements**

ANATOMICAL POSITION

- Starting reference point
- Specifies locations of specific body parts relative to other body parts



- Arms at the sides
- Palms facing forward
- Thumbs facing outward



- Standing erect
- Face forward

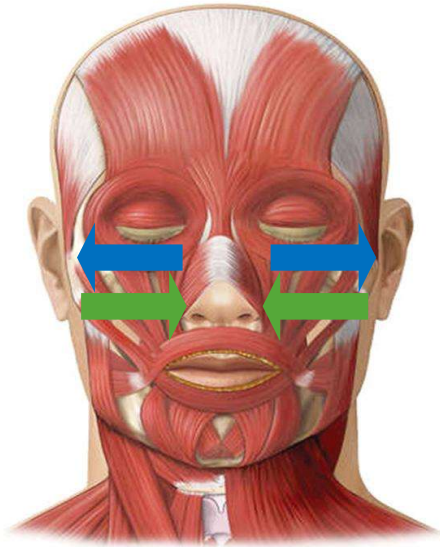
- Legs straight
- Heels & feet together, parallel

DIRECTIONAL TERMS

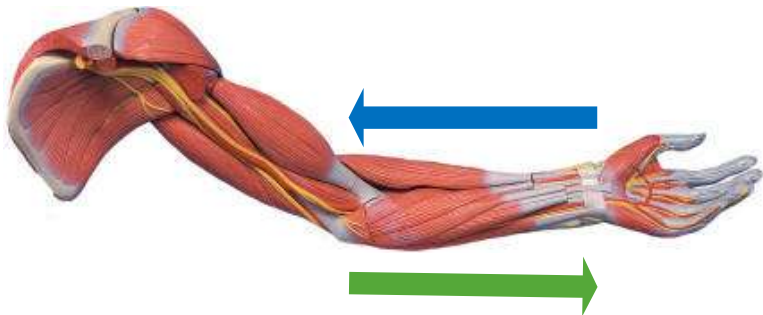
Based on the assumption that the body is in anatomical position



Ears **lateral** to cheeks
Cheeks **medial** to ears



Elbow **proximal** to hand
Hand **distal** to elbow



Anterior skeleton



Posterior skeleton



Pelvis **superior** to legs



Legs **inferior** to pelvis

Supine



Prone



_____ – Farth
The eyes are _____

Other examples?

_____ The quadric

_____ – Neare
The head is _____ thorax.

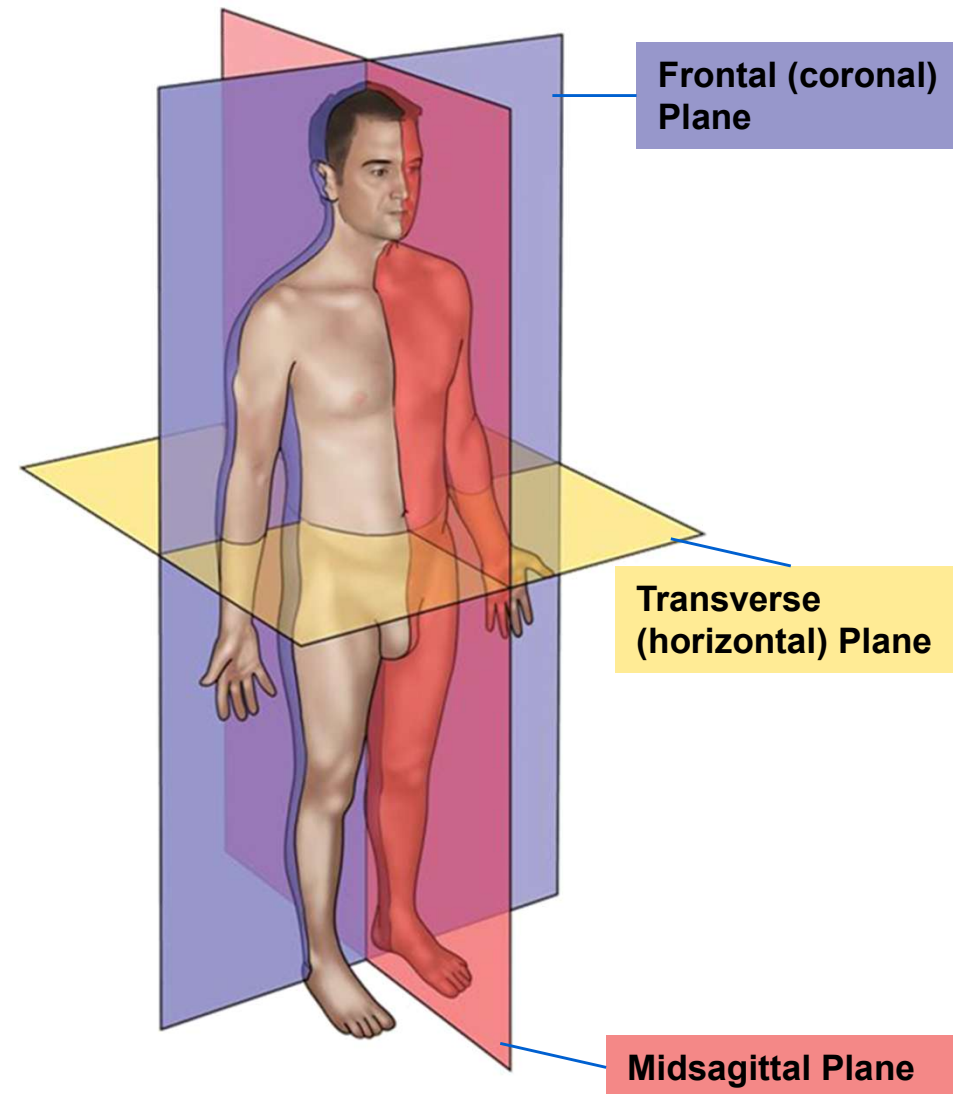
_____ – Nearer to the trunk
The arms are _____ to the hands.

_____ – Nearer to the median plane
The nose is _____ to the eyes.

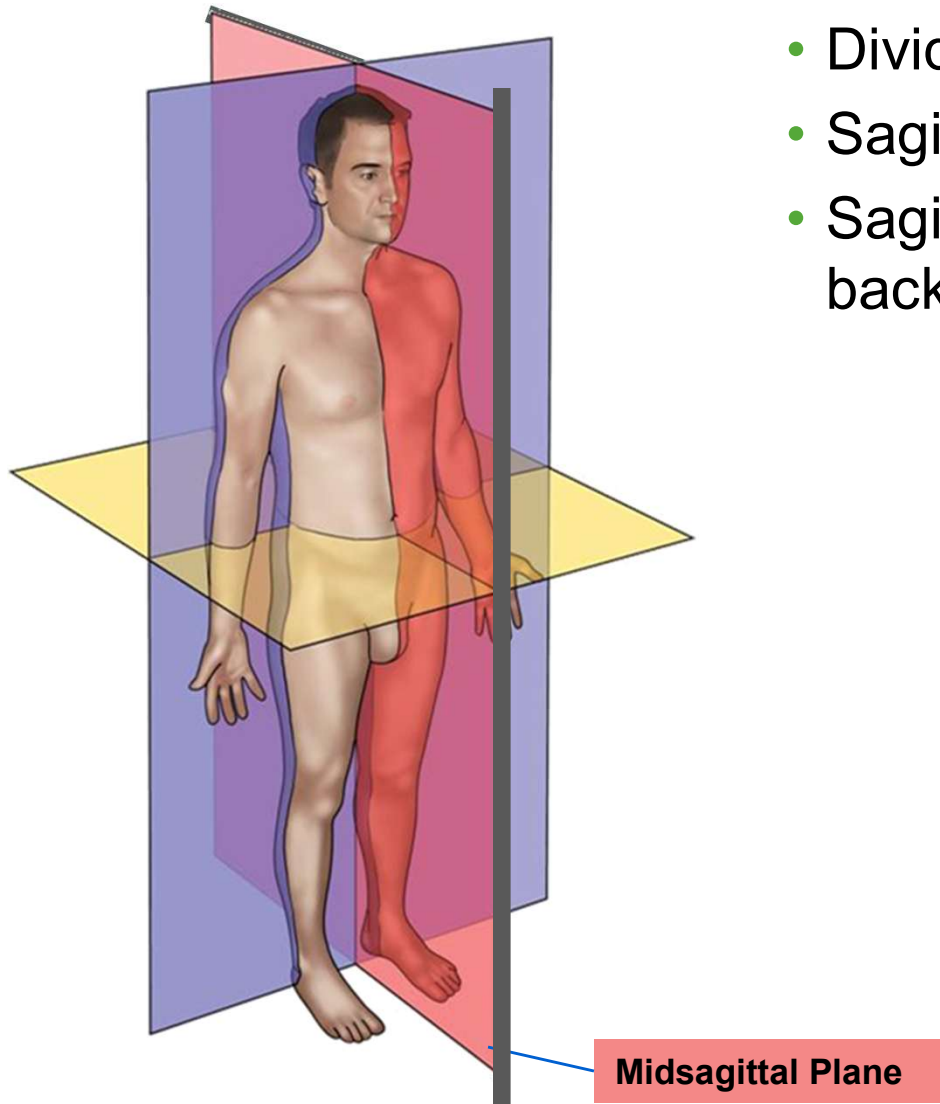
BODY PLANES

Imaginary flat surfaces:

- Divide the body in anatomical position
- At right angles to each other
- Describe movements in parallel to them



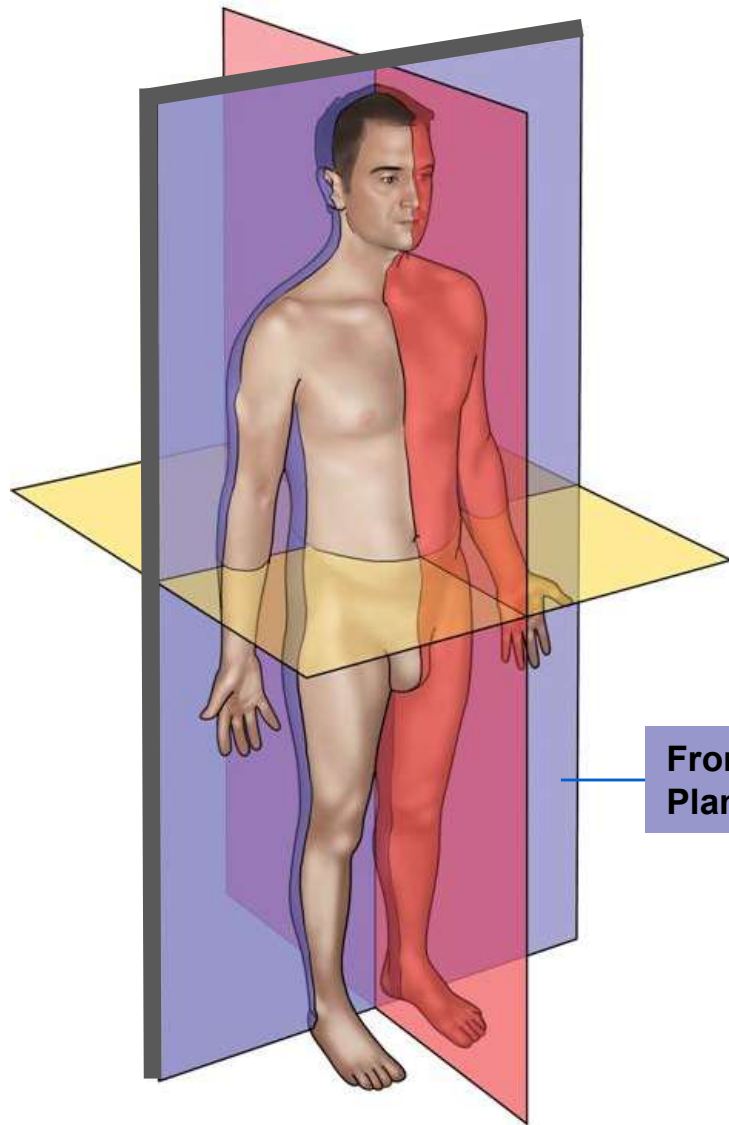
MIDSAGITTAL / MEDIAN PLANE



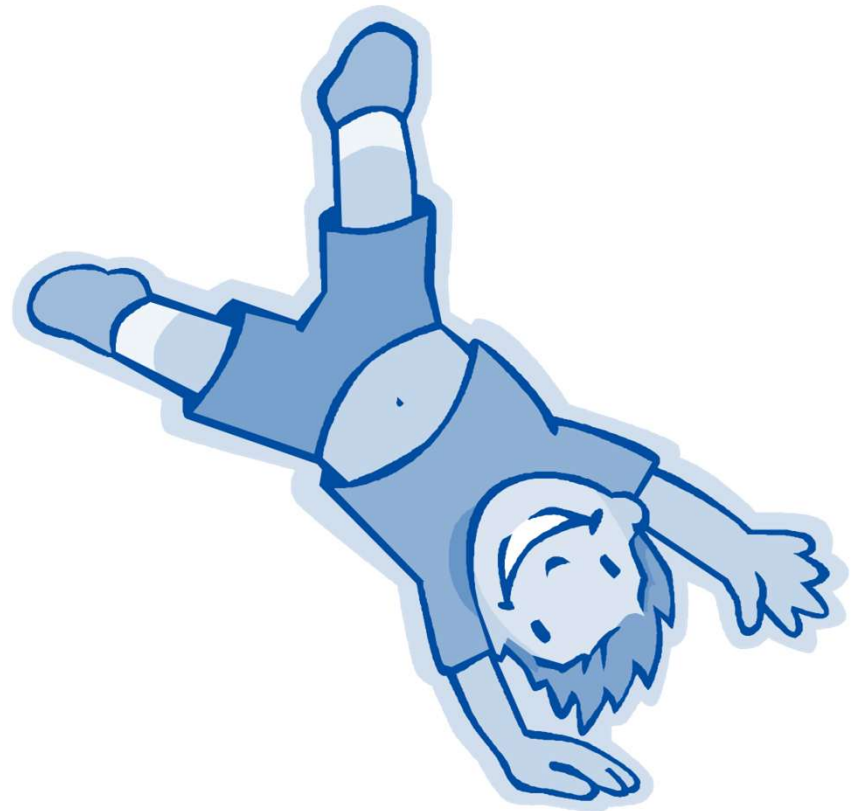
- Divides body into right and left halves
- Sagittal: parallel to median plane
- Sagittal plane movements: forward & backward motion



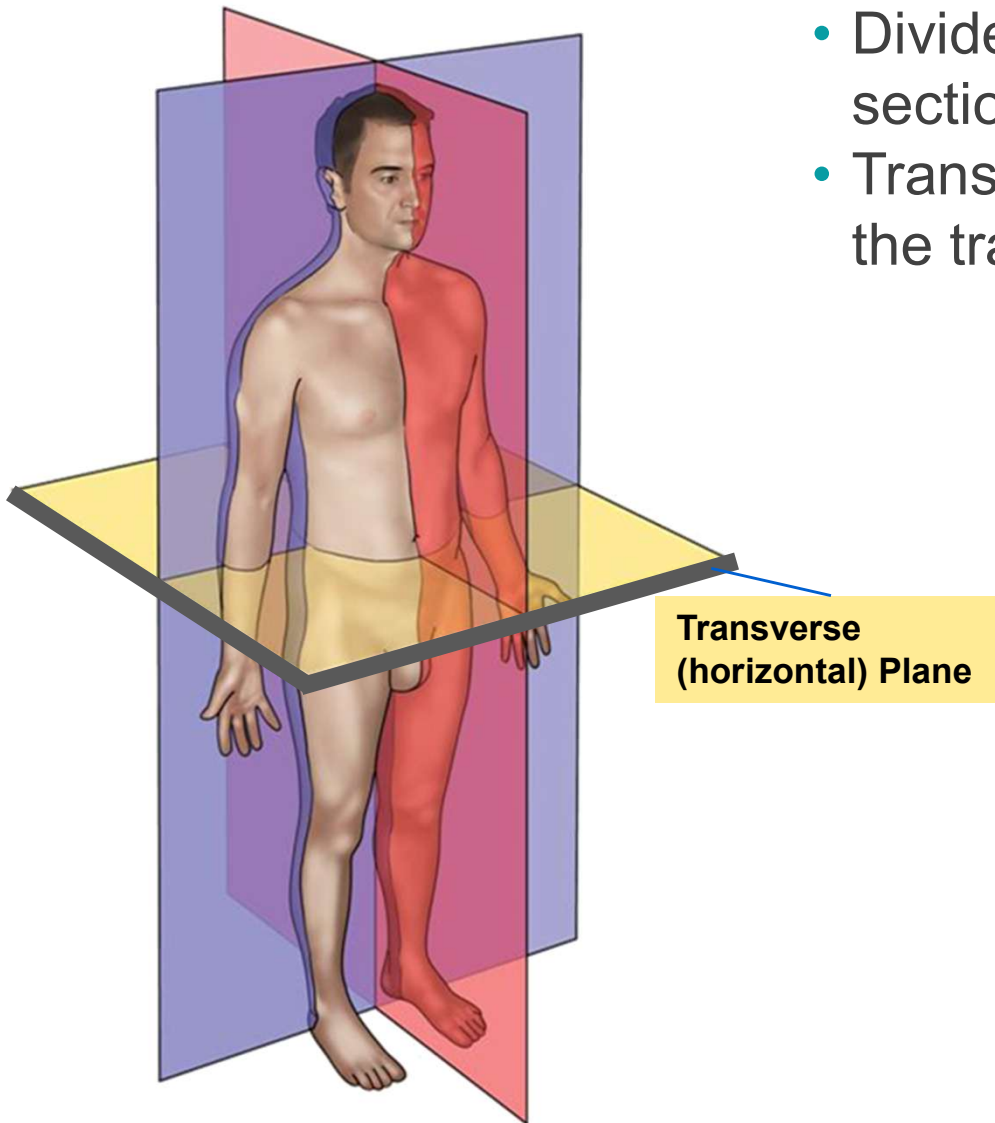
FRONTAL / CORONAL PLANE



- Divides body into anterior and posterior sections
- Frontal plane movements: lateral or side to side



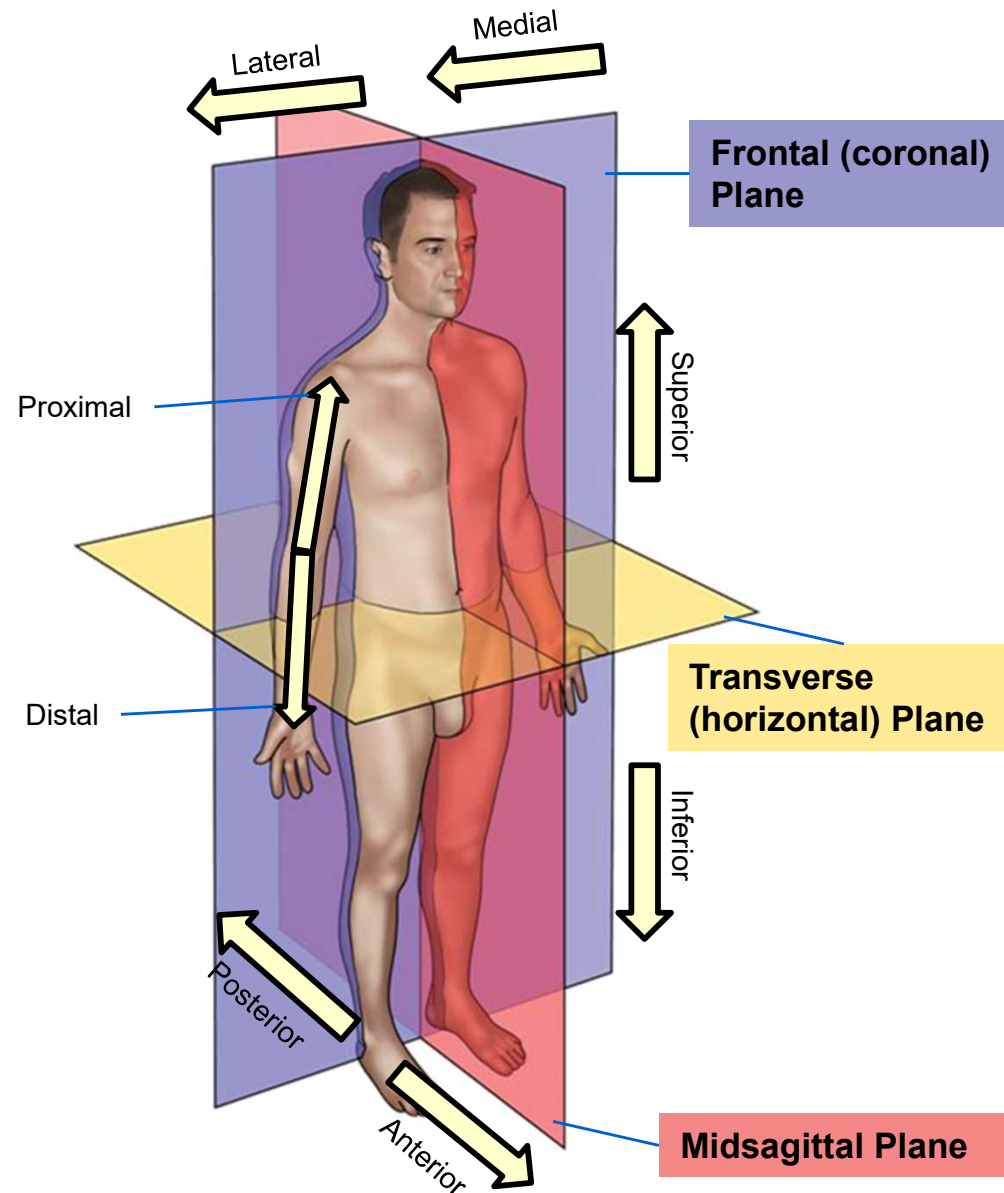
TRANSVERSE PLANE



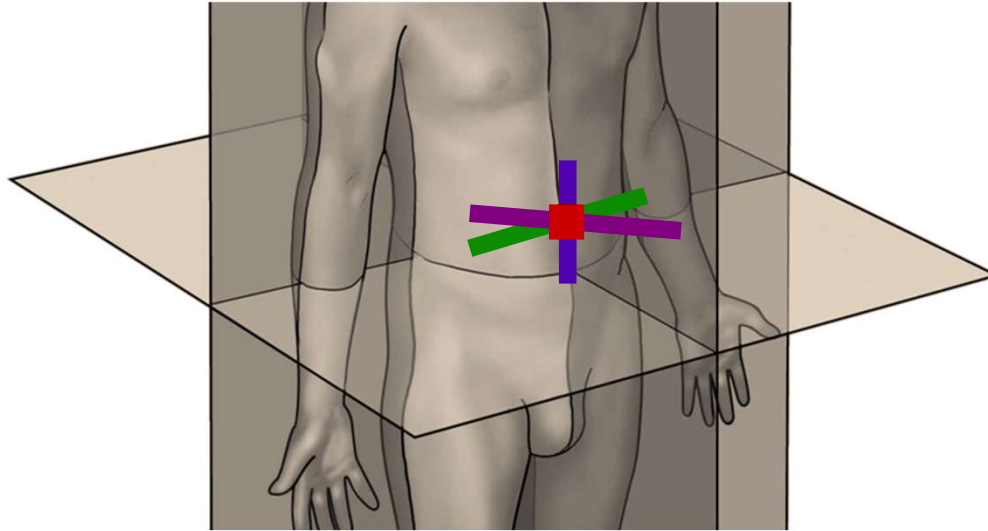
- Divides body into superior and inferior sections
- Transverse plane movements: parallel to the transverse plane



DIRECTIONAL TERMS



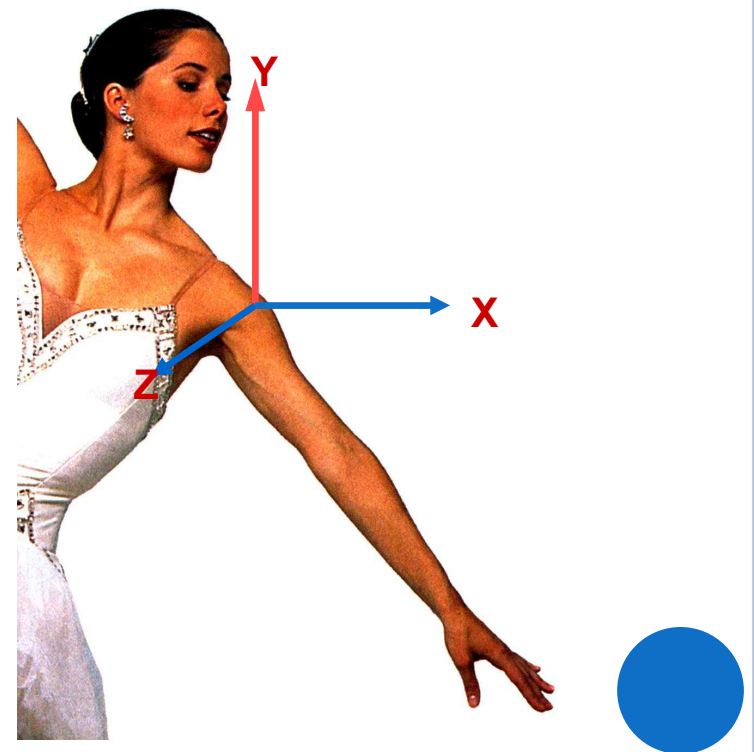
Center of gravity



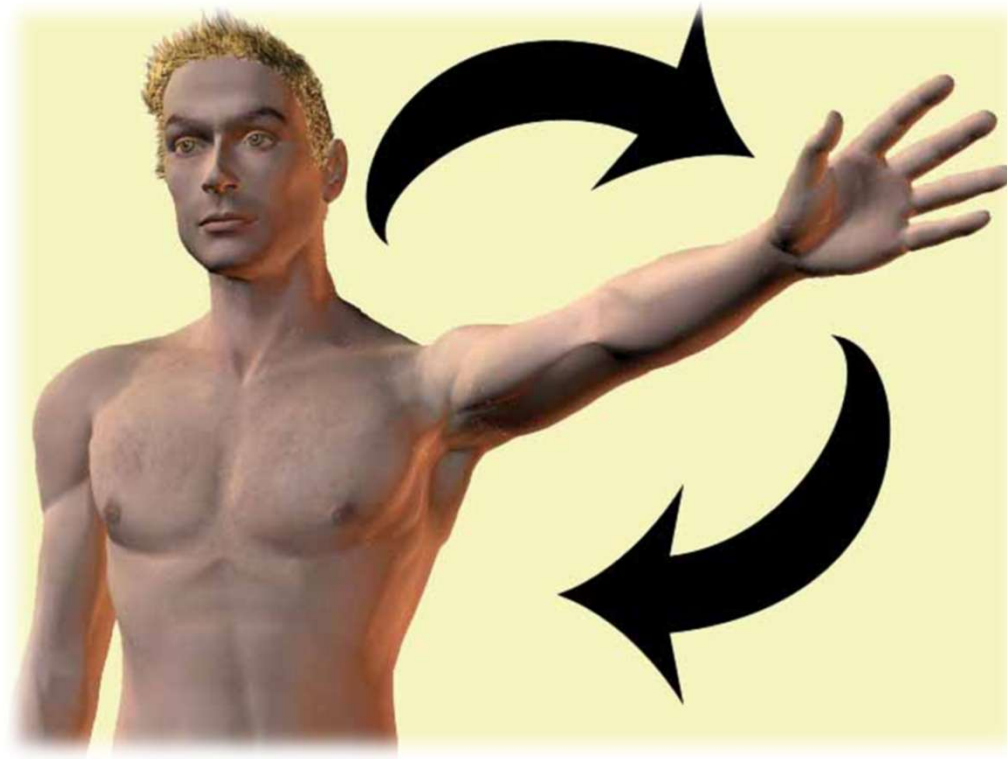
Where median, frontal, and transverse planes intersect

JOINT MOVEMENTS

- Around any one or more of three axes
- In pairs – opposite movements



Movement pair		Action	Reference	Example		
Flexion		Extension		Reduces / increases angle between joints	Sagittal plane movement	Biceps curl
Abduction		Adduction		Away / towards midline	Frontal plane movement	Jumping jacks
Pronation		Supination		Palm faces posterior / anterior	Hand and forearm movement	Holing bowl of soup
Dorsi-flexion		Plantar flexion		Foot up / foot plants	Sole foot movement	Toe raise
Inversion		Eversion		Sole moves in / out	Sole foot movement	Rolling over ankle
Medial / internal rotation		Lateral / external rotation		Flexed forearm moves in / out	Longitudinal axis movement	Opening / closing doors



CIRCUMDUCTION

- A cone of movement
 - No rotation
- Flexion/extension + abduction/adduction